Pacman AI — Output Analysis Report

1. Observed performance

A short one-episode comparison was run with three ghosts, a 20-step episode cap, and a fixed maze seed. The results are a sample, not a statistically reliable ranking; ghost placement and game behavior remain stochastic.

Agent	Kills	Score	Steps survived	Tiles travelled	Episode runtime	Timeouts
Minimax (depth 2)	3	30	20	20	1.318 s	0
Alpha-Beta (max depth 3)	2	20	20	18	2.401 s	0
MCTS (180 iterations, rollout 10)	3	30	20	20	0.459 s	0

On one five-ghost initial state, one observed decision took approximately 0.307 s for Minimax, 0.765 s for Alpha-Beta, and 0.166 s for MCTS. These are individual measurements rather than averages.

MCTS was the fastest in the short sample and tied for the most kills. Alpha-beta spent more time searching deeper and did not outperform Minimax in this single episode. More episodes and multiple seeds are needed before drawing a general performance conclusion. The `--assess` option runs all agents for a user-selected episode count and reports average kills, score, episode length, tiles travelled, and timeouts.